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United States Patent	12416208
Kind Code	B1
Date of Patent	September 16, 2025
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Robotic system for continuous circulation for a drilling operation

Abstract

A system can include a continuous circulation subcomponent, a robotic device, and a control system. The subcomponent can include a receiving area and can be associated with a drilling operation for forming a wellbore. The robotic device can include an end effector that can include a sensing device and tools corresponding to operations to facilitate continuous circulation in the wellbore. The control system can cause the robotic device to perform operations. The robotic device can align the end effector with the receiving area. The robotic device can determine whether a leak is present at the receiving area. The robotic device can connect, disconnect, or a combination thereof the diversion line with the receiving area to allow the continuous circulation during the drilling operation.

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Appl. No.: 18/784785

Filed: July 25, 2024

Publication Classification

Int. Cl.: E21B21/01 (20060101)

U.S. Cl.:

CPC E21B21/019 (20200501);

Field of Classification Search

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Background/Summary**TECHNICAL FIELD**

(1) The present disclosure relates generally to wellbore operations and, more particularly (although not necessarily exclusively), to a robotic system that can be used to facilitate continuous circulation

during a wellbore drilling operation.

BACKGROUND

(2) Wellbore operations may include various equipment, components, methods, or techniques to perform various tasks with respect to a wellbore, such as fluid control. In some examples, the wellbore operations may involve a drilling operation such as active wellbore drilling, tripping in, tripping out, and the like. During a drilling operation, mud and other fluid may be circulated, such as by pumping the mud or other fluid into a wellbore being formed, out of a wellbore being formed, etc., with respect to the wellbore being formed. As the wellbore is being formed, new piping or other material may be attached to a drill string or other component of the drilling operation to continue the drilling operation. Attaching the new piping or other material, for example without interrupting the drilling operation, may be difficult.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of a continuous circulation drilling system according to some aspects of the present disclosure.

(2) FIG. 2 is a block diagram of a robotic continuous circulation system that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure.

(3) FIG. 3 is a perspective view of a robotic continuous circulation system that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure.

(4) FIG. 4 is a bottom-perspective view of an end effector of a robotic continuous circulation system that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure.

(5) FIG. 5 is a block diagram of a computing system that can be used with a robotic continuous circulation system that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure.

(6) FIG. 6 is a flowchart of a process for using a robotic continuous circulation system to facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure.

DETAILED DESCRIPTION

(7) Certain aspects and examples of the present disclosure relate to a robotic system that can be used to facilitate continuous circulation for a drilling operation. The drilling operation can be or include an active drilling operation, a tripping in operation, a tripping out operation, or any other suitable operation involving a wellbore being formed. Continuous circulation can involve circulating fluid, such as mud or other material, into, out of, or within the wellbore without interrupting or otherwise stopping the drilling operation. For example, the continuous circulation can involve continuing to circulate the fluid while new piping or other new material is being coupled with or otherwise added to a drill string being used to form the wellbore. The robotic system can facilitate the continuous circulation. For example, the robotic system can include a robotic device, such as a robotic arm, and an end effector that can include a set of tools that can be used to enable the continuous circulation. Enabling the continuous circulation may involve diverting a source of the fluid circulating with respect to the wellbore from a main source to a diversion source that can circulate the fluid with respect to the wellbore and allow new piping or other new material to be coupled with or otherwise added to the drill string.

(8) Continuous circulation in a drilling operation can be a technique used, for example with respect to managed pressure drilling (MPD), to maintain flow down a drill pipe while making a connection, thereby maintaining equivalent circulating density (ECD) and keeping a constant pressure profile in well annulus to prevent an influx of formation fluids or potential hole collapse due to instability. In other drilling operations, the drilling fluid may not keep circulating in the

process of drill pipe connection, so the cuttings cannot be transported continuously. With continuous circulation, there may be no interruption to the flow of drilling fluid into the well throughout the process of adding drill pipe joints to, or removing drill pipe joints from, the drillstring. Continuous circulation drilling can provide a downhole, steady-state condition, and the formations may not suffer from pressure oscillations while hole-cleaning and borehole stability improve. Additionally or alternatively, the risk of drilling issues related to bottom hole pressure management in open holes can be mitigated or eliminated.

(9) A continuous circulation system can enable continuous circulation during a drill pipe connection process for drilling and tripping narrow-pressure window wells and providing continuous hole cleaning, among other suitable operations relating to drilling. The continuous circulation system can include a set of components, which may include a diversion manifold and a continuous circulation subcomponent. While the continuous circulation subcomponent can be integrated with a drillstring of the drilling operation, the manifold can be connected to a rig standpipe to divert flow from a top drive to a side port of the subcomponent through a diversion flow line, or vice versa. During continuous circulation, a set of steps can be performed to connect a diversion flow line from the manifold to the side port of the subcomponent and to disconnect after the circulation is done. Other systems for performing a subset of the set of steps rely on manual operations and expose operators and equipment to enhanced risk of damage, injury, and the like. The manual operations can involve connection and disconnection of the diversion flow line.

(10) A robotic continuous circulation system can be used to automatically connect and disconnect the diversion flow line to the side port of the subcomponent to allow a fully automated continuous circulation process. The robotic continuous circulation system can include a robotic device such as a robotic arm. During the continuous circulation process, the robotic device can connect or disconnect the diversion flow line from the manifold to or from the side port of the subcomponent. The robotic device can include one or more robotic arms that can be installed on the rig floor. Additionally or alternatively, each robotic arm can include an end effector that can have a manipulation toolset and a sensing device. The manipulation toolset can include a set of tools for manipulating the connection or disconnection of the diversion flow line, and the sensing device may include a camera or a position sensor that can be used to guide the end effector to the subcomponent.

(11) In some examples, using the robotic continuous circulation system can cause improvements to be made over other drilling systems. The improvements can include: maintaining formation integrity and borehole stability by reducing ballooning effects and pressure cycling on the well by removing pumps-on and pumps-off conditions. extending a life of a bottom-hole assembly of the drilling operation by maintaining the dynamic well temperature. reducing transitional errors during drill pipe connection by lessening the need for human intervention. preventing mud rheology or composition changes by maintaining a constant flow of fluid. reducing connection gas and any other negative side effects by stopping and restarting flow. improving the rate of penetration (ROP) through continuous hole cleaning.

(12) During continuous circulation, a first step may involve using the robotic continuous circulation system to connect a diversion flow line from the manifold to the side port of the subcomponent. After the circulation is complete, the robotic continuous circulation system can perform another step to disconnect the diversion flow line and resume the regular operation such as drilling or tripping. The above-mentioned steps may each have one or more sub-steps and may be automatically performed, such as without human intervention once initiated, in series such as with a set of tools positioned on the end effector of the robotic device.

(13) The robotic device of the robotic continuous circulation system can be integrated with a toolset and diversion flow line. For example, a particular tool of the toolset can be connected with the diversion flow line. The robotic device can bring the toolset, or the end effector that includes the toolset, near the subcomponent based on predefined position parameters, data gathered by the

sensing device, or a combination thereof. A control system can be included in the robotic continuous circulation system and can include a robot controller in communication with the robotic device to control the robotic device. The robotic continuous circulation system can make a connection and can disconnect or diversion flow line based on an analysis of image data to determine the location and orientation of the subcomponent, or any other component thereof, and to transmit the location and orientation of the subcomponent, or any other component thereof, to the robot controller.

(14) The toolset, which may be or may be included with the end effector, can include multiple tools to perform certain steps or operations related to connecting and disconnecting the diversion line with respect to the subcomponent. One tool included in the toolset may include a tool to loosen a bleeder valve to check a sealing condition of a flapper included in the subcomponent. The tool can also determine or can be used to determine if there is a leakage behind a safety plug of the subcomponent. Leakage or no leakage can be confirmed with data gathered, for example, by the sensing device. If there is no leakage, a second tool included with the toolset can be aligned to remove the safety plug and retain the safety plug. After removing the safety plug, the diversion flow line can be aligned and installed, for example by using a third tool included with the toolset. The diversion manifold can be operated to start the flow diversion. Upon completing the flow diversion, the above sequence of steps can be reversed to remove the diversion flow line, to install the safety plug, and to tighten the bleeder valve. In some examples, rotation movement by the robotic device operating the toolset can be driven by a motor associated with the robotic device. In some examples, each tool included in the toolset may be able to perform one operation or more than one operation. For example, one example of a tool can interact with a valve, can detect a leak or pressure, or a combination thereof.

(15) The sensing device can include one or more cameras on the toolset and may capture images, or other position data, of an object. The sensing device can identify features, such as hardware shape, position, etc., on the subcomponent and can determine the location and orientation of the features to manipulate the robotic arm and the toolset for engagement. While loosening the bleeder valve for leakage checking, techniques such as image comparison or motion detection can be used to determine if there is a leakage, to determine a magnitude of any detected leakage, and the like.

(16) Illustrative examples are given to introduce the reader to the general subject matter discussed herein and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

(17) FIG. 1 is a diagram of a continuous circulation drilling system **100** according to some aspects of the present disclosure. The continuous circulation drilling system **110** may include a derrick or drilling rig **120**, which may be located on land, for example as illustrated, or atop an offshore platform, semi-submersible, drill ship, or any other platform capable of forming a wellbore **113** through one or more downhole formations **111**. The continuous circulation drilling system **110** may be used in vertical wells, non-vertical or deviated wells, multilateral wells, offshore wells, etc.

(18) The continuous circulation drilling system **110** may include a top drive **124**, a hoist **126**, other equipment necessary for drilling the wellbore **113**, or any combination thereof. Additionally or alternatively with respect to the top drive **124**, a rotary table **128** may be provided. The drilling rig **120** may be located generally above a well head **114**, which in the case of an offshore location is located at the sea bed and may be connected to the drilling rig **120** via a riser or other suitable connection feature.

(19) The drilling rig **120** may be used to carry a drill string **132**, which may be assembled from individual lengths or stands of connected lengths of drill pipe **130** that may be run at least partially into the wellbore **113**. In some examples, the wellbore **113** may be completed or may be in a process of being formed). The drill string **132** may include standard drill pipe, heavy-wall drill

pipe, drill collars, coiled tubing, or any suitable combination thereof. The wellbore **113** may be at least partially lined with casing **119** along a length of the wellbore **113**. In some examples, the drill string **132** may include one or more continuous circulation subcomponents, such as the continuous circulation subcomponent **134**, along a length of the drill string **132**, which may have intervals between individual lengths or stands of the drill pipe **130**.

(20) A lower end of the drill string **132** may include a bottom hole assembly **150**, which may carry a rotary drill bit **152** at a distal end of the bottom hole assembly **150**. The bottom hole assembly **150** may include one or more drill collars, stabilizers, reamers, a downhole mud motor, rotary steerable device and various other tools such as those that can provide logging or measurement data and other information from the bottom of, or any other location with respect to, the wellbore **113**. Measurement data and other information may be communicated from bottom hole assembly **150** using measurement while drilling techniques and can be converted to electrical signals at the well surface **112** to, among other things, monitor the performance of the drill string **132**, the bottom hole assembly **150**, and the associated rotary drill bit **152**.

(21) An interior of the drill string **132** can define an axially extending conduit. An annulus **133** can be defined between the drill string **132** and a wall of the wellbore **113**. During drilling operations, a mud pump **148** may provide a drilling fluid **146** or other well treatment fluid such as weighted drilling mud, a cement slurry, a displacement fluid, a completion fluid, a stimulation fluid, a gravel pack fluid, and the like, from a mud pit **140**, through the interior of drill string **132**, through the bottom hole assembly **150**, to exit through nozzles within rotary drill bit **152**. The drilling fluid **146** may then mix with formation cuttings and other downhole fluids and debris. The annulus **133** may provide a flow path for the drilling fluid to be returned to the mud pit **140** at the surface **112**. Various types of screens, filters, centrifuges, and the like may be provided to remove formation cuttings and other downhole debris prior to returning drilling fluid to recirculation by the mud pump **148**.

(22) The continuous circulation subcomponent **134** can allow the drilling fluid **146** to be circulated through the wellbore **113** to continue without interruption during the operational steps for making or breaking connections of the drill string **132** at the rig floor. In some examples, the continuous circulation subcomponent **134** can include a tubular body with pin and box connectors **135** and **136** at opposite ends for connection along the drill string **132**. Additional or alternative and suitable connector types may be used for the drill string **132**. In some examples, the continuous circulation subcomponent **134** may be a unitary subcomponent or may be an assembly of discrete subcomponents. In a particular example, the continuous circulation drilling system **110** may use an individual continuous circulation subcomponent connected atop each drill pipe **130** as the stand is being made up to or removed from the drill string **132**. The tubular body of the continuous circulation subcomponent **134** may define an axial flow path between connectors **135** and **136**. An axial flow valve can be disposed within the continuous circulation subcomponent **134**, and the axial flow valve can act as a one-way check valve that allows flow in one direction such as a downhole direction. In some examples, the axial valve may be a swing check flapper valve, although other types of valves may be used as appropriate.

(23) The continuous circulation drilling system **110** may operate to maintain continuous circulation in the wellbore **113**. For example, and during drilling operations, which may include an active drilling operation, a tripping operation, etc., the drill string **132** can be lowered into wellbore **113** via a hoist **126** and rotated by the top drive **124** or the rotary table **128**. Drilling fluid can be supplied by the mud pump **148** via a flow line **142**, a flow manifold **144**, a hose **145**, and the top drive **124** or a fluid swivel to the axial flow path through a top connector, such as connector **136** of the continuous circulation subcomponent **134**. An axial valve of the continuous circulation subcomponent **134** can be opened, and a radial valve of the continuous circulation subcomponent **134** can be shut. Fluid can flow out of a bottom connector, such as the connector **135**, into the interior of the drill string **132**.

(24) Lowering and rotating the drill string **132** may be temporarily ceased when the continuous circulation subcomponent **134** reaches the level of the drilling rig floor. The drill string **132** may be held by slips within the rotary table **128**. A robotic continuous circulation system **200** can be used with respect to the continuous circulation subcomponent **134** at the elevation of side port **137**. The robotic continuous circulation system **200** may, for example using an end effector that includes a set of different tools, automatically check the leakage within the side port **137** between a bleeder valve and a safety plug, may automatically remove the safety plug from the side port **137**, and may automatically attach a diversion flow line **199** into the side port **137**. The flow manifold **144** may then be operated to divert drilling fluid flowing through the hose **145** and the connector **136** into the axial flow path of the continuous circulation subcomponent **134** to a hose **147** and the diversion flow line **199** into the side port **137** of the continuous circulation subcomponent **134**. The pressure differential within the continuous circulation subcomponent **134** can operate to shut the axial valve and open the radial valve within continuous circulation subcomponent **134**. The robotic continuous circulation system **200** can be used to disconnect the diversion flow line **199** from the continuous circulation subcomponent **134** once the drill string **132** has been augmented with an additional drill pipe or for other suitable reasons. The robotic continuous circulation system **200** can disconnect the diversion flow line **199** using the same or similar tools with respect to connecting the diversion flow line **199** to the continuous circulation subcomponent **134**.

(25) FIG. 2 is a block diagram of a robotic continuous circulation system **200** that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure. As illustrated in FIG. 2, the robotic continuous circulation system **200** can include the flow manifold **144**, a robotic device **202**, the continuous circulation subcomponent **134**, and a control system **204**. The robotic continuous circulation system **200** may include additional, alternative, or fewer components.

(26) The continuous circulation subcomponent **134** may be positioned on the drill string **132** that can be run in the wellbore **113** for forming the wellbore **113** or for other suitable purposes. The continuous circulation subcomponent **134** may include a receiving area **206**, such as the side port **137**, that can be sized to receive a diversion line **208** such as the diversion flow line **199**. In some examples, the receiving area **206** may include a safety plug, a bleeder valve, and other components to facilitate continuous circulation, and associated operations, for a drilling operation involving the drill string **132**. The bleeder valve can allow leaks to be detected in the continuous circulation subcomponent **134** without causing damage to the drill string **132** or the robotic continuous circulation system **200**. The safety plug may be removed to allow the diversion line **208** to be attached, such as via the robotic device **202**, to the continuous circulation subcomponent **134** at the receiving area **206**. In some examples, the diversion line **208** may extend from the flow manifold **144** to the robotic device **202**, or any component thereof such as the end effector **212** or any tools of the set of tools **214** positioned on the end effector **212**.

(27) The robotic device **202** can be or include a robotic arm **210** that can include a SCARA robot, a Cartesian robot, and the like. In some examples, the robotic device **202** can include the robotic arm **210** and an end effector **212** that can be positioned on a distal end of the robotic arm **210** to allow the robotic device **202** to interact with the continuous circulation subcomponent **134** or other suitable components associated with the drill string **132**. The robotic device **202** may include the control system **204**, may be communicatively coupled with the control system **204**, or may be in any other suitable configuration with respect to the control system **204**. As illustrated, the robotic device **202** includes the control system **204**, but in other examples, the control system **204** may be separated from the control system **204** and may be communicatively coupled with the control system **204** to allow the control system **204** to control the robotic device **202**. In some examples, controlling the robotic device **202** can involve determining a trajectory for moving the robotic device **202**, determining a force applied by the robotic device **202**, control of a motor for moving or adjusting an orientation of the end effector **212**, or any tools thereof, and the like.

(28) The end effector **212** may be positioned on an end of the robotic device **202** proximate to the continuous circulation subcomponent **134** when the continuous circulation subcomponent **134** is positioned on a rig floor to facilitate continuous circulation. In some examples, the end effector **212** can include a set of tools **214** and a sensing device **216**, though the end effector **212** can include any additional, alternative, or fewer components to provide functionality for the robotic device **202**. The set of tools **214** can include more than one tool to allow the end effector to perform more than one operation without changing hardware or a configuration of the robotic device **202**. In some examples, each tool included in the set of tools **214** may be capable of performing one operation or more than one operation. In a particular example, the set of tools **214** can include three tools that can each interact with, or otherwise be positioned at, the receiving area **206** of the continuous circulation subcomponent **134**. The three tools can include a first tool for checking for a leak in the continuous circulation subcomponent **134**, can include a third tool for removing a safety plug to access the receiving area **206** of the continuous circulation subcomponent **134**, and can include a second tool coupled with the diversion line **208**. The second tool can be connected to the receiving area **206** to at least temporarily affix the diversion line **208** to the receiving area **206** of the continuous circulation subcomponent **134** to allow fluid to flow from the diversion line **208** into the continuous circulation subcomponent **134** and into the wellbore **113**.

(29) The sensing device **216** can be or include a camera, a laser sensor, a position sensor, or other suitable device that can record and transmit position data about the end effector **212**, the receiving area **206** of the continuous circulation subcomponent **134**, or a combination thereof. For example, the sensing device **216** can be a camera that can record locations of the end effector **212** and the receiving area **206** of the continuous circulation subcomponent **134** while the robotic continuous circulation system **200** is in operation. The sensing device **216** may provide recorded data to the control system **204** that can use the data from the sensing device **216** to control the robotic device **202**, for example by adjusting a trajectory or motor speed associated with the robotic device **202**.

(30) In some examples, the robotic continuous circulation system **200** can be used to facilitate a continuous circulation operation with respect to a drilling operation using the drill string **132**. The robotic continuous circulation system **200** can perform a set of operations to determine that the continuous circulation subcomponent **134** is safe to open, to access the continuous circulation subcomponent **134**, and to connect the diversion line **208** to the continuous circulation subcomponent **134**. By connecting the diversion line **208** to the continuous circulation subcomponent **134**, a section of new piping **218** can be added to the drill string **132** to allow the drill string **132** to go further into the wellbore **113** and to avoid pausing fluid circulation with respect to the wellbore **113**.

(31) FIG. 3 is a perspective view of a robotic continuous circulation system **200** that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure. As illustrated in FIG. 3, the robotic continuous circulation system **200** can include the robotic device **202**, the flow manifold **144**, the control system **204**, and the continuous circulation subcomponent **134**. The robotic continuous circulation system **200** may include additional, alternative, or fewer components than are illustrated in FIG. 3.

(32) The robotic continuous circulation system **200** can be used to divert flow of fluid in the wellbore **113** using the flow manifold **144**. For example, the diversion line **208** can be affixed to the continuous circulation subcomponent **134** by the robotic continuous circulation system **200** to divert fluid from being received upstream of the drill string **132** to being received via the diversion line **208**, for example to allow additional pipe sections to be added to the drill string **132** or for other suitable purposes. The robotic continuous circulation system **200** may use the robotic device **202** to perform a set of operations for connecting the diversion line **208** to the continuous circulation subcomponent **134** or for disconnecting the diversion line **208** from the continuous circulation subcomponent **134**.

(33) The robotic device **202** may include the end effector **212**, which may include the set of tools

214 and the sensing device **216**. The robotic continuous circulation system **200** can gather data about continuous circulation subcomponent **134**, such as a location of the receiving area **206**, a type of hardware used to seal the receiving area **206**, etc., using the sensing device **216** and can provide the gathered data to the control system **204** to control the robotic device **202**. Controlling the robotic device **202** can involve controlling each tool of the set of tools **214** or any subset thereof. As illustrated in FIG. 3, the set of tools **214** can include a first tool **302a**, a second tool **302b**, and a third tool **302c**. The first tool **302a** can be or include a tool to detect pressure or leaks that may be present in the continuous circulation subcomponent **134**. The third tool **302c** may be or include a tool for removing a safety plug from the continuous circulation subcomponent **134** to provide access to the continuous circulation subcomponent **134** such as via the receiving area **206**. The second tool **302b** may be or include a tool that can be connected to the diversion line **208** and that can affix the diversion line **208** to the receiving area **206** of the continuous circulation subcomponent **134**. The first tool **302a**, the second tool **302b**, and the third tool **302c** may be or include any other suitable tool for facilitating operations to affix the diversion line **208** to the continuous circulation subcomponent **134** or to otherwise facilitate continuous circulation in the wellbore **113**.

(34) FIG. 4 is a bottom-perspective view of an end effector **212** of a robotic continuous circulation system **200** that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure. The end effector **212** may be coupled with the robotic device **202** and may be controlled by the control system **204** or other suitable control module. Additionally or alternatively, each tool illustrated on the end effector **212** may be powered via an electrical motor, a pneumatic motor, or other suitable motors that can provide power to the tools of the end effector **212**. For example, each tool included on the end effector **212** may have a different motor or a distinct but similar motor to provide functionality to the tool.

(35) As illustrated in FIG. 4, the end effector **212** can include the set of tools **214** that can include the first tool **302a**, the second tool **302b**, and the third tool **302c**, though the set of tools **214** may include any other suitable number (e.g., less than three or more than three) of tools. In some examples, the first tool **302a**, the second tool **302b**, the third tool **302c**, or any combination thereof, can be or include a screwdriver, wrench, a ratchet, or other tool that can provide functionality for the robotic continuous circulation system **200**. The tools included in the set of tools **214** may be specially designed or adapted for use on the end effector **212**. For example, if the first tool **302a** is a wrench, the first tool **302a** may be a modified wrench head that does not include a handle and that simply includes a wrench head sized to interact with a specific component of the continuous circulation subcomponent **134**.

(36) As illustrated in FIG. 4, the set of tools **214** are spaced apart horizontally from one another in a configuration in which the end effector **212** is interacting with the receiving area **206**. For example, the set of tools **214** may be arranged, from left to right, as the first tool **302a**, the second tool **302b**, and then the third tool **302c**. Other configurations, such as vertically spaced apart, radially arranged, and other configurations, are possible. Additionally or alternatively, the sensing device **216** may be positioned above the set of tools **214** in a configuration in which the end effector **212** is interacting with the receiving area **206**. The sensing device **216** may be otherwise suitably arranged with respect to the set of tools **214** to provide sensing functionality for the robotic continuous circulation system **200** or any component thereof.

(37) In some examples, the diversion line **208** can be coupled with at least one tool of the set of tools **214**. As illustrated in FIG. 4, a first end **402** of the diversion line **208** can be coupled with the second tool **302b**, though in other examples, the diversion line **208** can be coupled with other tools of the set of tools **214**. The first end **402** can be connected to the receiving area **206** to at least temporarily affix the diversion line **208** to the continuous circulation subcomponent **134**. Additionally or alternatively, prior to at least temporarily affixing the diversion line **208** to the continuous circulation subcomponent **134**, a first end **404** of the first tool **302a** and a first end **406**

of the third tool **302c** may be used to prepare the receiving area **206** of the continuous circulation subcomponent **134** for the first end **402**. For example, the first end **404** can be used to check whether a leak exists in the continuous circulation subcomponent **134** by opening a bleed valve, and the first end **406** can be used to remove a safety plug from the receiving area **206** to provide a surface for the first end **402** to couple with the continuous circulation subcomponent **134**. Other operations and configurations for the set of tools, and ends thereof, can be performed or used with respect to the end effector **212**.

(38) FIG. 5 is a block diagram of a computing system **500** that can be used with a robotic continuous circulation system **200** that can facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure. The components, such as processor **504**, memory **507**, power source **520**, input/output **508**, and so on, illustrated in FIG. 5, may be integrated into a single structure such as within a single housing of the control system **204** and in communication with the robotic device **202**. In some examples, the components illustrated in FIG. 5 can be distributed from one another and may be in electrical communication with each other. And, in other examples, the computing system **500** may be integrated into the robotic device **202**, or vice versa.

(39) The control system **204** can include the processor **504**, the memory **507**, and a bus **506**, among other suitable components for the control system **204**. The processor **504** can execute one or more operations for performing a set of operations for facilitating continuous circulation in the wellbore **113**. The processor **504** can execute computer-program instructions **510** stored in the memory **507** to perform the set of operations. The processor **504** can include one processing device or multiple processing devices or cores. Non-limiting examples of the processor **504** can include a field-programmable gate array (“FPGA”), an application-specific integrated circuit (“ASIC”), a microprocessor, and the like.

(40) The processor **504** can be communicatively coupled with the memory **507** via the bus **506**. The memory **507** may be or include non-volatile memory and may include any type of memory device that retains stored information when powered off. Some examples of non-volatile forms of the memory **507** may include EEPROM, flash memory, or any other type of non-volatile memory. In some examples, at least part of the memory **507** can include a medium from which the processor **504** can read computer-program instructions **510**. A computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processor **504** with computer-readable instructions or other program code. Some examples of a computer-readable medium may include magnetic disk(s), memory chip(s), ROM, RAM, an ASIC, a configured processor, optical storage, or any other medium from which a computer processor can read computer-program instructions **510**. The computer-program instructions **510** can include processor-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, including, for example, C, C++, C#, Perl, Java, Python, etc.

(41) In some examples, the memory **507** can be a non-transitory computer readable medium and can include computer-program instructions **510**. The computer-program instructions **510** can be executed by the processor **504** for causing the processor **504** to perform the set of operations. For example, the processor **504** can execute a robot control service **511**, and the like to provide functionality for the control system **204**, the robotic device **202**, or the like. For example, the processor **504** can cause sensor data **513**, such as position data, to be gathered by the robotic device **202** such as via the sensing device **216**. Additionally or alternatively, the processor **504** can execute a robot motor service **512** to cause the robotic device **202**, or any component thereof such as the end effector **212**, to be translated or otherwise moved to align the end effector **212** with the continuous circulation subcomponent **134**, or the receiving area **206** thereof. Additionally or alternatively, the processor **504** can execute the robot control service **511** to perform a set of operations for causing the robotic device **202** to interact with the continuous circulation subcomponent **134** for facilitating continuous circulation in the wellbore **113**.

(42) The control system **204** can additionally include an input/output **508**. The input/output **508** can connect to a keyboard, a pointing device, a display, other computer input/output devices or any combination thereof. An operator may provide input using the input/output **508**. In some examples, the control system **204** may be fully autonomous and may function without input from an operator. Data relating to the wellbore **113**, the continuous circulation subcomponent **134**, the robotic continuous circulation system **200**, the drilling operation, or any combination thereof can be displayed to an operator of a wellbore operation through a display that is connected to or is part of the input/output **508**. The displayed values can be observed by the operator, or by another suitable user, of the wellbore operation, who can adjust the wellbore operation based on the output.

Additionally or alternatively, the control system **204** can automatically control or adjust the wellbore operation, which may be or include a drilling operation, based on the output.

(43) FIG. **6** is a flowchart of a process **600** for using a robotic continuous circulation system **200** to facilitate continuous circulation for a drilling operation according to some aspects of the present disclosure. At block **602**, an end effector **212** is aligned with a receiving area **206** of a continuous circulation subcomponent **134**. The end effector **212** may be positioned on or otherwise coupled with the robotic device **202** and may include a set of tools **214** and a sensing device **216**. The end effector **212** may be aligned by the control system **204** and based on data gathered by the sensing device **216**. In some examples, the control system **204** may cause the end effector **212** to be positioned within a predetermined area proximate to an expected location of the continuous circulation subcomponent **134**. After the end effector **212** is positioned in the predetermined area, the sensing device **216** may gather position data, image data, or other suitable data about the end effector **212**, about the continuous circulation subcomponent **134**, or the receiving area **206** thereof, or any combination thereof. The sensing device **216** can provide the gathered data to the control system **204**, which can use the gathered data to make one or more fine adjustments to an alignment of the end effector **212**. The fine adjustments to the end effector **212** may cause the end effector **212**, or the set of tools **214** thereof, to be aligned with the receiving area **206**. In some examples, the end effector **212** being aligned with the receiving area **206** may involve the end effector **212** being in a position to interact or engage with the receiving area **206** to perform a set of operations with the set of tools **214**.

(44) At block **604**, a first tool of the set of tools **214** is used to engage with a valve positioned at the receiving area **206**. In some examples, the first tool can engage with the valve by physically interlocking with the valve and rotating the valve a predetermined amount to perform a first particular operation. The first particular operation may include determining whether a leak is present at the receiving area **206** or otherwise within the continuous circulation subcomponent **134**. The valve may be or include a bleeder valve that, if a leak is present, may apply pressure on pressure sensor adjacent to the first tool or may otherwise be detected by the sensing device **216**. In response to determining that a leak is present, the control system **204** may cause the first tool to tighten the valve and to take no further action until the continuous circulation subcomponent **134** is repaired or decommissioned. In some examples, if a leak is detected, a next continuous circulation subcomponent may be positioned on which for the robotic continuous circulation system **200** to operate. In response to detecting that no leak is present, the control system **204** may proceed in the process **600** by using a third tool to perform a second operation.

(45) At block **606**, the third tool of the set of tools **214** is used to engage with and remove a safety plug from the continuous circulation subcomponent **134** to provide access to the receiving area **206**. In some examples, the third tool may be offset, such as horizontally offset, vertically offset, radially offset, etc., from the first tool, and the control system **204** may cause the end effector **212** to horizontally, vertically, or radially translate to align the third tool with the receiving area **206** or the safety plug installed thereon. In some examples, engaging the third tool with the safety plug may involve physically interlocking an end of the third tool with the safety plug and rotating the safety plug to remove the safety plug. In other examples, the safety plug can be removed through

other interactions, such as lifting upward, pulling outward, etc., other than rotation. In response to removing the safety plug, the third tool may retain or otherwise hold on to the safety plug until the safety plug is used later in the process **600** or is otherwise used in other suitable processes.

(46) At block **608**, a second tool is used to connect the diversion line **208** with the receiving area **206**. In some examples, the second tool may be offset, such as horizontally offset, vertically offset, radially offset, etc., from the first tool, from the third tool, or from a combination thereof. The control system **204** may cause the end effector **212** to horizontally translate, vertically translate, radially translate, or any combination thereof to cause the second tool to be aligned with the receiving area **206**. In some examples, the second tool may be coupled with the diversion line **208**. In a particular example, a first end of the second tool may be or include a female-type connector or a male-type connector that is sized to be affixed to the receiving area **206**, and a second end of the second tool opposite the first end may be integrated with or otherwise coupled with the diversion line **208**. Affixing the second tool to the receiving area **206**, such as by screwing the second tool onto the receiving area **206**, pressing the second tool into the receiving area **206**, or other suitable mechanisms for affixing the second tool to the receiving area **206**, can at least indirectly affix the diversion line **208** to the receiving area **206**. For example, affixing the second tool to the receiving area **206** may allow fluid to flow from the diversion line **208**, through the receiving area **206**, and into the wellbore **113**, for example without losing the fluid.

(47) The diversion line **208** may be retained in the receiving area **206** for a predetermined amount of time or other suitable amount of time corresponding to an operation associated with continuous circulation in the wellbore **113**. For example, the diversion line **208** may be retained as affixed to the receiving area **206** for an amount of time to allow additional pipe to be added upstream from the continuous circulation subcomponent **134** to the drill string **132**.

(48) At block **610**, and in response to determining that an operation associated with continuous circulation in the wellbore **113** is complete, the above operations are performed in reverse. For example, the control system **204** may cause the second tool, and the diversion line **208**, to be disconnected from the receiving area **206**. Additionally or alternatively, the control system **204** may align the third tool with the receiving area **206** and may cause the third tool to reattach the safety plug to the receiving area **206**. Additionally or alternatively, the control system **204** may align the first tool with the receiving area **206** and may cause the first tool to tighten the valve to allow the continuous circulation subcomponent **134** to receive fluid from upstream for circulation in the wellbore **113**. In some examples, after removing at least the diversion line **208** from the receiving area **206**, the robotic device **202** can clean, such as by spraying with detergent or water, the receiving area **206** or other portions of the continuous circulation subcomponent **134**. Additionally or alternatively, the robotic device **202** can be moved to an initial position to await further instructions from the control system **204**.

(49) In some aspects, systems, methods, and robotic devices for continuous circulation for drilling operations are provided according to one or more of the following examples:

(50) As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”). Example 1 is a system comprising: a continuous circulation subcomponent comprising a receiving area for a diversion line for continuous circulation during a drilling operation involving a wellbore; a robotic device comprising one or more end effectors, the one or more end effectors comprising a sensing device and a plurality of tools corresponding to a plurality of operations to facilitate the continuous circulation; and a control system communicatively coupled with the robotic device to cause the robotic device to perform operations comprising: aligning at least one end effector of the one or more end effectors with the receiving area; determining whether a leak is present at the receiving area; and in accordance with a determination that the leak is not present, connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation. Example 2 is the system of example 1, wherein the sensing device comprises a

camera, a laser sensor, or a position sensor, and wherein the operation of aligning the at least one end effector with the receiving area comprises using position data from the camera, the laser sensor, or the position sensor to align the at least one end effector with the receiving area. Example 3 is the system of example 1, wherein the plurality of tools comprises a first tool, a second tool, and a third tool, wherein the first tool is arranged on the at least one end effector to perform a first operation of the plurality of operations, wherein the second tool is arranged on the at least one end effector to perform a second operation of the plurality of operations, wherein the third tool is arranged on the at least one end effector to perform a third operation of the plurality of operations, and wherein the first operation, the second operation, and the third operation are different from one another. Example 4 is the system of any of examples 1 or 3, wherein the first tool, the second tool, and the third tool are horizontally offset from one another when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent, and wherein the sensing device is vertically offset from the plurality of tools when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent. Example 5 is the system of any of examples 1 or 3, wherein the first operation is the operation of determining whether the leak is present at the receiving area, wherein the third operation is removing a safety plug from the receiving area, and wherein the second operation is the operation of connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation. Example 6 is the system of any of examples 1, 3, or 5, wherein the second tool is coupled with the diversion line, and wherein the second tool is couplable with the receiving area of the continuous circulation subcomponent to connect the diversion line with the continuous circulation subcomponent to allow the continuous circulation with respect to the wellbore. Example 7 is the system of example 1, wherein the operation of connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation comprises retaining the at least one end effector at the receiving area while the diversion line is being used to facilitate the continuous circulation. Example 8 is the system of any of examples 1 or 7, wherein the operations further comprise, in response to determining that the continuous circulation is complete, removing the diversion line from the receiving area with the at least one end effector. Example 9 is a method comprising: aligning, by a control system of a robotic continuous circulation system that comprises a robotic device, at least one end effector of the robotic device with a receiving area of a continuous circulation subcomponent associated with a drilling operation to form a wellbore, the aligning based on data received from a sensing device positioned on the at least one end effector; determining, by the control system and using a first tool of a plurality of tools included in one or more end effectors that comprise the at least one end effector, whether a leak is present at the receiving area of the continuous circulation subcomponent; and in accordance with a determination that the leak is not present, and in response to using a third tool to access the receiving area, connecting, by the control system and using a second tool of the plurality of tools, a diversion line with the receiving area to allow continuous circulation in the wellbore during the drilling operation. Example 10 is the method of example 9, wherein the sensing device comprises a camera, a laser sensor, or a position sensor, and wherein aligning the at least one end effector with the receiving area comprises using, by the control system, position data from the camera, the laser sensor, or the position sensor to align the at least one end effector with the receiving area. Example 11 is the method of example 9, wherein the plurality of tools comprises the first tool, the second tool, and the third tool, wherein the first tool is arranged on the at least one end effector to perform a first operation of a plurality of operations, wherein the second tool is arranged on the at least one end effector to perform a second operation of the plurality of operations, wherein the third tool is arranged on the at least one end effector to perform a third operation of the plurality of operations, and wherein the first operation, the second operation, and the third operation are different from one another. Example 12 is the method of any of examples 9 or 11, wherein the first operation is determining whether the leak is present at the receiving area, wherein the second operation is

connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation, and wherein the third operation is removing a safety plug from the receiving area to access the continuous circulation subcomponent. Example 13 is the method of any of examples 9 or 11-12, wherein the second tool is coupled with the diversion line, and wherein connecting the diversion line with the receiving area comprises affixing the second tool to the receiving area to couple the diversion line with the receiving area. Example 14 is the method of example 9, wherein connecting the diversion line with the receiving area comprises retaining the at least one end effector at the receiving area while the diversion line is being used to facilitate the continuous circulation. Example 15 is the method of any of examples 9 or 14, further comprising, in response to determining that the continuous circulation is complete, removing the diversion line from the receiving area with the at least one end effector. Example 16 is a robotic device comprising: one or more end effectors comprising a sensing device and a plurality of tools corresponding to a plurality of operations to facilitate continuous circulation in a wellbore for a drilling operation; and a control system to cause the robotic device to perform operations comprising: aligning at least one end effector of the one or more end effectors with a receiving area of a continuous circulation subcomponent associated with the drilling operation; determining whether a leak is present at the receiving area; and in accordance with a determination that the leak is not present, connecting a diversion line with the receiving area to allow the continuous circulation during the drilling operation. Example 17 is the robotic device of example 16, wherein the sensing device comprises a camera, a laser sensor, or a position sensor, and wherein the operation of aligning the at least one end effector with the receiving area comprises using position data from the camera, the laser sensor, or the position sensor to align the at least one end effector with the receiving area. Example 18 is the robotic device of example 16, wherein the plurality of tools comprises a first tool, a second tool, and a third tool, wherein the first tool is arranged on the at least one end effector to perform a first operation of the plurality of operations, wherein the second tool is arranged on the at least one end effector to perform a second operation of the plurality of operations, wherein the third tool is arranged on the at least one end effector to perform a third operation of the plurality of operations, and wherein the first operation, the second operation, and the third operation are different from one another. Example 19 is the robotic device of any of examples 16 or 18, wherein the first tool, the second tool, and the third tool are horizontally offset from one another when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent, and wherein the sensing device is vertically offset from the plurality of tools when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent. Example 20 is the robotic device of example 16, wherein the operation of connecting the diversion line with the receiving area comprises retaining the at least one end effector at the receiving area while the diversion line is being used to facilitate the continuous circulation, and wherein the operations further comprise, in response to determining that the continuous circulation is complete, removing the diversion line from the receiving area with the at least one end effector.

(51) The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. A system comprising: a continuous circulation subcomponent comprising a receiving area for a diversion line for continuous circulation during a drilling operation involving a wellbore; a robotic device comprising one or more end effectors, the one or more end effectors comprising a sensing

device and a plurality of tools corresponding to a plurality of operations to facilitate the continuous circulation; and a control system communicatively coupled with the robotic device to cause the robotic device to perform operations comprising: aligning at least one end effector of the one or more end effectors with the receiving area; determining whether a leak is present at the receiving area; and in accordance with a determination that the leak is not present, connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation.

2. The system of claim 1, wherein the sensing device comprises a camera, a laser sensor, or a position sensor, and wherein the operation of aligning the at least one end effector with the receiving area comprises using position data from the camera, the laser sensor, or the position sensor to align the at least one end effector with the receiving area.

3. The system of claim 1, wherein the plurality of tools comprises a first tool, a second tool, and a third tool, wherein the first tool is arranged on the at least one end effector to perform a first operation of the plurality of operations, wherein the second tool is arranged on the at least one end effector to perform a second operation of the plurality of operations, wherein the third tool is arranged on the at least one end effector to perform a third operation of the plurality of operations, and wherein the first operation, the second operation, and the third operation are different from one another.

4. The system of claim 3, wherein the first tool, the second tool, and the third tool are horizontally offset from one another when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent, and wherein the sensing device is vertically offset from the plurality of tools when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent.

5. The system of claim 3, wherein the first operation is the operation of determining whether the leak is present at the receiving area, wherein the third operation is removing a safety plug from the receiving area, and wherein the second operation is the operation of connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation.

6. The system of claim 5, wherein the second tool is coupled with the diversion line, and wherein the second tool is couplable with the receiving area of the continuous circulation subcomponent to connect the diversion line with the continuous circulation subcomponent to allow the continuous circulation with respect to the wellbore.

7. The system of claim 1, wherein the operation of connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation comprises retaining the at least one end effector at the receiving area while the diversion line is being used to facilitate the continuous circulation.

8. The system of claim 7, wherein the operations further comprise, in response to determining that the continuous circulation is complete, removing the diversion line from the receiving area with the at least one end effector.

9. A method comprising: aligning, by a control system of a robotic continuous circulation system that comprises a robotic device, at least one end effector of the robotic device with a receiving area of a continuous circulation subcomponent associated with a drilling operation to form a wellbore, the aligning based on data received from a sensing device positioned on the at least one end effector; determining, by the control system and using a first tool of a plurality of tools included in one or more end effectors that comprise the at least one end effector, whether a leak is present at the receiving area of the continuous circulation subcomponent; and in accordance with a determination that the leak is not present, and in response to using a third tool to access the receiving area, connecting, by the control system and using a second tool of the plurality of tools, a diversion line with the receiving area to allow continuous circulation in the wellbore during the drilling operation.

10. The method of claim 9, wherein the sensing device comprises a camera, a laser sensor, or a position sensor, and wherein aligning the at least one end effector with the receiving area comprises

using, by the control system, position data from the camera, the laser sensor, or the position sensor to align the at least one end effector with the receiving area.

11. The method of claim 9, wherein the plurality of tools comprises the first tool, the second tool, and the third tool, wherein the first tool is arranged on the at least one end effector to perform a first operation of a plurality of operations, wherein the second tool is arranged on the at least one end effector to perform a second operation of the plurality of operations, wherein the third tool is arranged on the at least one end effector to perform a third operation of the plurality of operations, and wherein the first operation, the second operation, and the third operation are different from one another.

12. The method of claim 11, wherein the first operation is determining whether the leak is present at the receiving area, wherein the second operation is connecting the diversion line with the receiving area to allow the continuous circulation during the drilling operation, and wherein the third operation is removing a safety plug from the receiving area to access the continuous circulation subcomponent.

13. The method of claim 12, wherein the second tool is coupled with the diversion line, and wherein connecting the diversion line with the receiving area comprises affixing the second tool to the receiving area to couple the diversion line with the receiving area.

14. The method of claim 9, wherein connecting the diversion line with the receiving area comprises retaining the at least one end effector at the receiving area while the diversion line is being used to facilitate the continuous circulation.

15. The method of claim 14, further comprising, in response to determining that the continuous circulation is complete, removing the diversion line from the receiving area with the at least one end effector.

16. A robotic device comprising: one or more end effectors comprising a sensing device and a plurality of tools corresponding to a plurality of operations to facilitate continuous circulation in a wellbore for a drilling operation; and a control system to cause the robotic device to perform operations comprising: aligning at least one end effector of the one or more end effectors with a receiving area of a continuous circulation subcomponent associated with the drilling operation; determining whether a leak is present at the receiving area; and in accordance with a determination that the leak is not present, connecting a diversion line with the receiving area to allow the continuous circulation during the drilling operation.

17. The robotic device of claim 16, wherein the sensing device comprises a camera, a laser sensor, or a position sensor, and wherein the operation of aligning the at least one end effector with the receiving area comprises using position data from the camera, the laser sensor, or the position sensor to align the at least one end effector with the receiving area.

18. The robotic device of claim 16, wherein the plurality of tools comprises a first tool, a second tool, and a third tool, wherein the first tool is arranged on the at least one end effector to perform a first operation of the plurality of operations, wherein the second tool is arranged on the at least one end effector to perform a second operation of the plurality of operations, wherein the third tool is arranged on the at least one end effector to perform a third operation of the plurality of operations, and wherein the first operation, the second operation, and the third operation are different from one another.

19. The robotic device of claim 18, wherein the first tool, the second tool, and the third tool are horizontally offset from one another when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent, and wherein the sensing device is vertically offset from the plurality of tools when the at least one end effector is connected with the receiving area of the continuous circulation subcomponent.

20. The robotic device of claim 16, wherein the operation of connecting the diversion line with the receiving area comprises retaining the at least one end effector at the receiving area while the diversion line is being used to facilitate the continuous circulation, and wherein the operations

further comprise, in response to determining that the continuous circulation is complete, removing the diversion line from the receiving area with the at least one end effector.
